IOWA STATE UNIVERSITY

Knowledge-Grounded 3D Disease Editing: What Does 3D Add Beyond 2D?

Tirtho Roy

Ushashi Bhattacharjee, Adarsh Krishnamurthy, Soumik Sarkar
Iowa State University, Ames, Iowa, USA

Pilot results and three focused tests of what 3D adds
Advisor update

October 2026
Ames, Iowa

Photo: Beardshear Hall, Tony Webster, CC BY-SA 2.0

The Question We Have to Answer

What does knowledge-guided 3D give us that the original 2D photo, or a knowledge-guided 2D edit, does not?

01 **Not enough: 2D → 3D → 2D**
Rendering a reconstruction back to the input view adds nothing the photo did not have.

02 **Candidate gains**
Shape symptoms (curl, pucker, galls), hidden surfaces (leaf underside), views a camera never took, one consistent object across views.

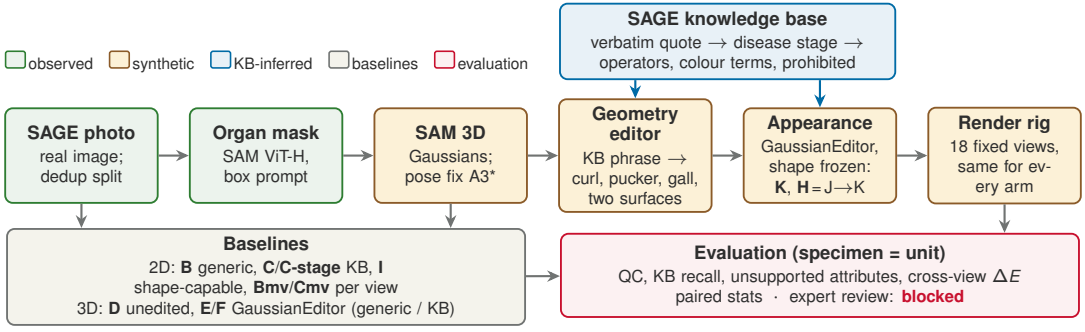
03 **Our rule**
Every claim must be measured, and the measurement must say whether it is *real*, *synthetic* or *unverified*.

How this deck is organised

Act I. Question, five hypotheses, pipeline, status.
Act II. Protocol: audit, case selection, data tiers, structured KB, matched arms.
Act III. Experiments E1–E5, three focused tests of what 3D adds, ablations, statistics, failures.
Act IV. Go / no-go, answers to the 12 questions, next steps.

Status. Pilot runs complete on all 15 specimens; three focused tests added. Numbers carry their *n* and are regenerated from the result files (2026-10-14 17:09).

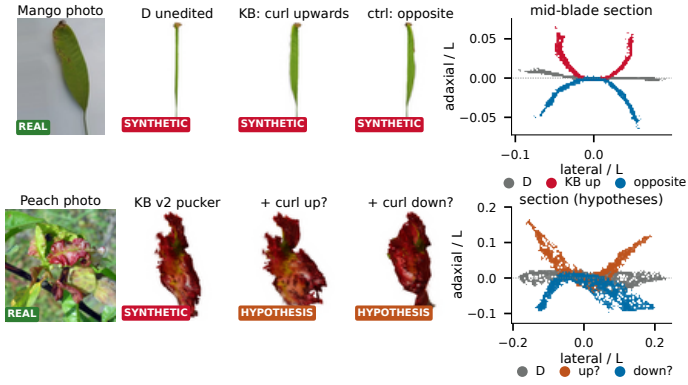
Pipeline as Implemented



Two editors. GaussianEditor changes colour, not shape (E2); shape comes from our *geometry editor* (KB terms without an operator are logged, never approximated). Three streams stay separate: observed (kept from the photo), KB-inferred (a cited phrase), synthetic (filled by a model, unverified).

OUR RUN – MEASURED diagram traced from the code paths (src/figures/fig1_pipeline.py lists them). SAM 3D Objects, GaussianEditor (Chen *et al.*, CVPR 2024), InstructPix2Pix (Brooks *et al.*, CVPR 2023).

Test 1: Directional Shape From the Text



Positive control: the operator follows a direction

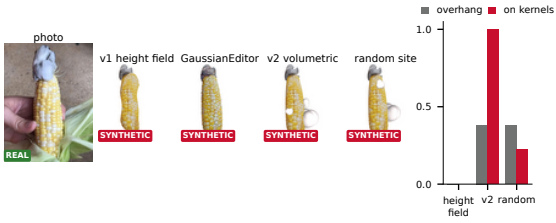
Counterfactual (leaves have anthracnose): "curl upwards" raises margins 3/3 (4.8° →46.2°), reversed 3/3 (-35.4°). IoU 0.91→0.62: contradicts the photo. Verifies code, not anatomy.

Peach: the source gives no direction

The operator **abstains**; up/down only as labelled hypotheses. Pucker on reddish tissue: enrichment 1.14 vs 1.07 uniform.

SYNTHETIC – UNVERIFIED **OUR RUN – MEASURED** results/demos/{mango_curl,peach}/*/metrics.json; roll = signed margin angle about the midrib (+ = toward the upper side). Recovering an imposed angle is a compliance check; a real curl phenotype is still needed.

Test 2: Galls as Volumes on Kernel Tissue



What a height field cannot do

v1 only pushes the surface out (no overhang). v2 galls are closed lobed bodies; 0.38 of their surface overhangs the ear.

Placement from evidence

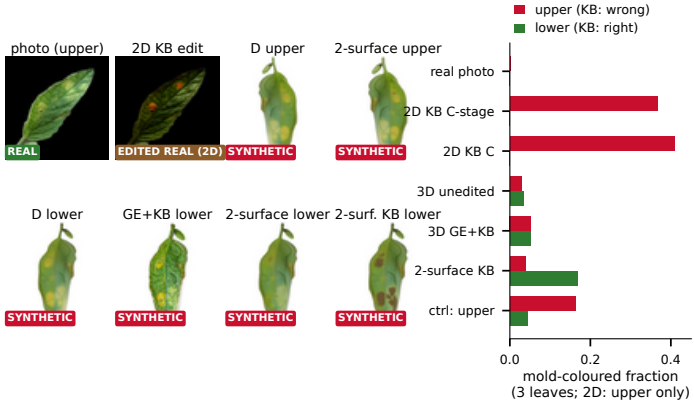
"...develop on the ear replacing kernels" (UNL): on kernel tissue in 1.00 of runs vs 0.22 random: *constraint compliance*, not a finding.

What is assumed

Ear length 18 cm (no ruler); count and position are in no source. 1 ear abstained (no kernel tissue).

REAL SAGE DATA **SYNTHETIC – UNVERIFIED** **OUR RUN – MEASURED** results/demos/corn/*/metrics.json, n = 2 ears × 3 seeds. Added volume 6.1 vs 4.5‰ of (ear length)^3 (galls vs height field; scale-free); 35.7 vs 26.4 cm^3 only under the assumed 18 cm ear. Not validated against real galls.

Test 3: Upper vs Lower Leaf Surface



2D puts the mold on the wrong side

KB: mold "on the lower leaf surface below leaf spots". A photo shows the upper side only; KB 2D edits paint mold colour on 17–49% (C-stage) / 31–53% (C) of it; real: 0–0%.

Two surfaces keep it apart

One sheet mirrors upper spots below (12–20% of lower pixels). Two surfaces: mold 7–29% of lower pixels, upper 0–8% (baseline 0–7%); 73–84% of it under real spots (random 7–23%).

REAL SAGE DATA | SYNTHETIC – UNVERIFIED | OUR RUN – MEASURED — results/demos/tomato/*/metrics.json, 3 leaves × 3 seeds, all 18 rig views. Colour classes fixed before the edits. The real underside is *unknown*: this shows the KB statement can be represented, not that it is right.

Ablations: Knowledge × Geometry

2×2 on total KB recall

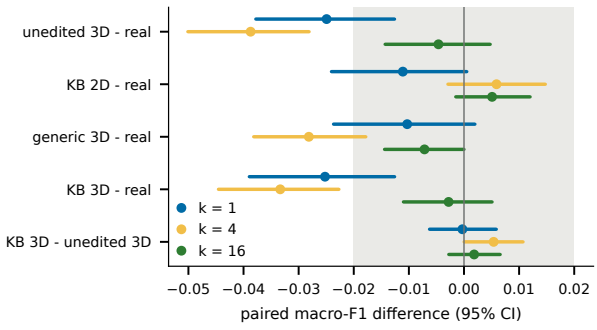
	no KB	KB
no geometry	D 0.00	K 0.17
geometry	G 0.04	H 0.53

Interaction $(H-G)-(K-D) = +0.32$, **n = 6** specimens with every cell (2 classes).

Ablation	Arms	Status
No KB / generic	B, E, G	✓
KB, wrong stage	C-NG	✓
Mismatched KB (neg. ctrl)	C-MM	✓
KB appearance only	K	✓
KB geometry only	J	✓
Appearance vs geometry 3D	K vs J vs H	✓
Placement: evidence vs random	J2/J2u, V/Vr, BLk/BLr	✓
Direction: cited vs opposite	CU/CD	✓
Viewpoints (1/2/4/8/adaptive)	–	✗
Severity, occlusion, background	–	✗

OUR RUN – MEASURED results/statistics/ablation_2x2.json. Total KB recall = mean of geometric and colour recall over the stage's required attributes.

E5: Minimal Downstream Test on Real Photos

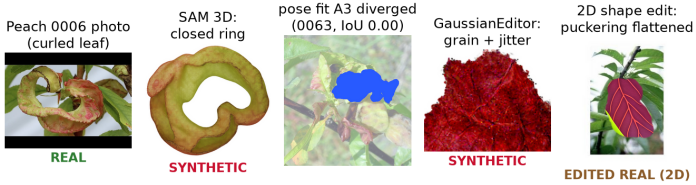


k real/class	KB-3D − real only	KB-3D − unedited 3D
1	-0.025	-0.000
4	-0.033	+0.005
16	-0.003	+0.002

Verdict (pre-registered): NOT SUPPORTED (falsified at this scale). The knowledge edit adds under 2 macro-F1 points over unedited renders; renders themselves hurt at small *k*. The render-to-photo gap comes before knowledge.

OUR RUN – MEASURED reports/14_downstream_minimal.md: within-crop disease classification, 5 crops, frozen DINOv2 + linear probe, 50 paired draws per *k*, 2,936 held-out real SAGE photos, pHash-disjoint. Macro-F1 differences; 95% CIs in results/downstream_minimal/contrasts.csv.

Failure Taxonomy



Reconstruction

Ring topology; 13/15 over-scaled; one-sheet leaves.

→ ✓ pose fix A3*; ✓ two-surface leaf (Test 3); ● reject closed rings before editing.

Optimisation

Pose refinement diverged on 3/15.

→ ✓ accept-if-better rule; ● add held-out depth as a second gate (worse on 6/15).

Editing

GE: grain, jitter, no shape; J blisters at random.

→ ✓ geometry editor for shape; ✓ evidence placement (weak gain); ● better lesion detector.

Biology

2D edit removes real puckering; stage mixing.

→ ✓ stage-aware KB; ● observed-morphology constraint; expert ratings blocked.

REAL SAGE DATA | SYNTHETIC – UNVERIFIED reports/09_failure_analysis.md; 00_full_failure_audit.md (root cause and status of each). ✓ done, ● next.

Execution Plan: Validate Before Generating More

01 Blinded plant-pathologist ratings
84-item package ready; 2–3 raters; one primary outcome declared first.

02 Small matched multi-view set
Corn smut and Tomato Leaf Mold first, with a ruler; held-out views score surface, direction, volume.

03 Independent geometry evaluation
Reference curl, gall volume and surface localisation, not the operators' own detectors.

04 Render-to-real diagnosis
Small matched pilot (background, lighting, texture, pose) before any 250 GPU-h E5.

05 Stronger baselines
KB-constrained 2D warp, two-layer 2D, shared-texture multi-view 2D.

Blocker	Unblocked by
No raters	3 recruited experts (01)
No matched real views	specimen capture (02)
No physical scale	ruler in captures
Render-to-photo gap	diagnosis (04)
Peach curl direction	a verbatim source

Question for you. Can we fund about 45 person-hours of capture and recruit three raters? Both precede any validity claim.

OUR RUN – MEASURED order follows the external review (validation first). Full blocker list and deliverables: backup slides; reports/12–15.

Go / No-Go After the Pilot

Decision	Required evidence vs what we have	Verdict
GO: KB-grounded 3D reconstruction	Verified hidden-trait recovery and a measured gain over KB 2D. *No held-out real views exist; minimal downstream test: no gain.*	✗ not met
PIVOT: 3D simulation and inspection	Plausible, controllable shape edits with limited claims. *Surface, direction and volume are representable and measured; consistent views; not yet expert-rated.*	● **best fit**
PIVOT: 2D knowledge-guided editing	Knowledge helps, 3D adds nothing. *Knowledge does steer 2D colour; but 2D cannot represent shape correctly (arm I).*	● partly
NO-GO: unreliable synthetic biology	Errors dominate. *Failures are real (ring, GE jitter) but diagnosed, reproducible and avoidable.*	✗ not dominant

Recommendation. Narrow the claim to *knowledge-controlled 3D symptom simulation*: an auditable KB→shape editor whose edits are traceable to cited text and consistent across views. Do not claim reconstruction of hidden anatomy. Numerical go/no-go thresholds will be set from pilot variances, not before.

OUR RUN – MEASURED evidence from slides E1–E4; decision logic from reports/10_final_experimental_protocol.md (Outcome D for now: no established advantage on real data).